Exam Solution – Applied Artificial Intelligence (DV 2557)

30th October, 2020

Max 33 pts: E 17-19, D 20-23, C 24-28, B 29-31, A 32-33 pts

Please note:

- *All the questions should be answered in English.*
- *The exam is closed book and the use of any book, notes or internet is not allowed.*

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1) What according to Daniel Gilbert and Jeff Hawkins is the right definition of intelligence? (2 pts)

According to Daniel Gilbert, the fundamental difference between humans and the animals is that the human mind can think of the future. Jeff Hawkins also reinforced this concept and believed that intelligence is not defined by behavior but prediction (which humans can do).

2) How is a goal-based agent different from utility-based agent? Can you implement a goal-based agent using a utility-based agent and if yes, how? (3 pts)

A goal-based agent executes actions that leads it closer to specific goals. A utility-based agent on the other hand executes its action by moving to states carrying best or higher utilities. For ex, both goal-based agent and utility agents can take you from A to B but a utility-based agent would do it in a more efficient manner like through the shortest path. It is possible to realize a goal-based agent using a utility agent. One way is by setting a high utility for the goal state.

3) Which method gives the best of breadth-first search and depth-first search? Explain its working how with the help of a (short) figurative example. (3 pts)

Iterative deepening gives the best of breadth-first and depth-first search methods. It is relatively fast and has low memory requirement like DFS. However, it saves you from getting stuck in a deep branch by setting the maximum depth and increasing depth iteratively. Explain with a figure.

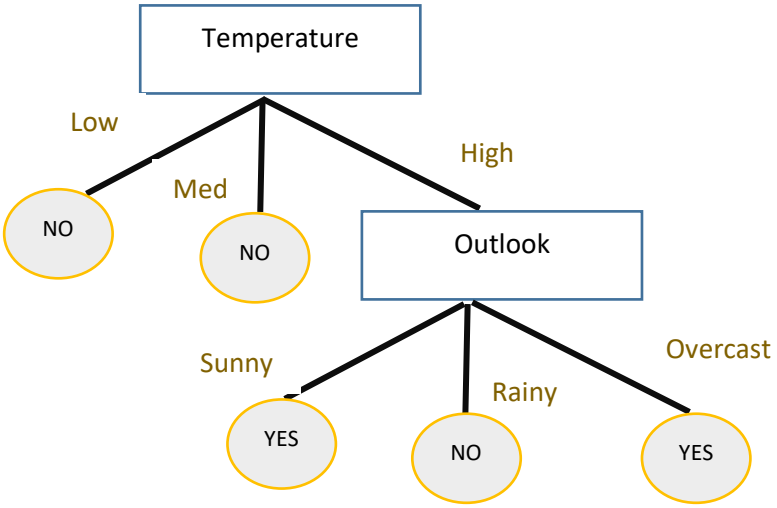
4) What is "min-conflicts" approach in context of local constraint satisfaction problem? State the most useful property of solving problems through min-conflicts. (3 pts)

Min-conflicts is a local search algorithm for CSP. Given a number of variables to which values are to be assigned such that certain constraint(s) need to be respected. The algorithm proceeds by assigning random value to each variable. In each iteration, a random variable is selected to be updated. This variable is updated such that it causes least number of conflicts with its neighbors locally. We repeat these iterations till we are done or reach a maximum number of iterations.

The most useful property of min-conflicts algorithm is that the time taken to solve a problem is roughly independent of the size of the problem and it can solve hard problems. For example, solving for various sizes (N) of N-queen problem.

5) Create a Decision Tree for the following dataset (GoForDrive is the class attribute): (4 pts)

Temperature = Low, Outlook = Sunny → GoForDrive = NO
Temperature = Medium, Outlook = Sunny → GoForDrive = NO
Temperature = High, Outlook = Sunny → GoForDrive = YES
Temperature = Medium, Outlook = Rainy → GoForDrive = NO
Temperature = High, Outlook = Rainy → GoForDrive = NO
Temperature = High, Outlook = Overcast → GoForDrive = YES
Temperature = Low, Outlook = Overcast → GoForDrive = NO
Temperature = Medium, Outlook = Overcast → GoForDrive = NO



** More than one correct trees possible. Any of those will be considered right.*

6) What are genetic algorithms? Explain briefly all steps involved in a genetic algorithm. (3 pts)

Genetic Algorithms (GA) are a class of optimization algorithms where varied population evolved independently is mixed together to obtain a solution. GA consist of the following major steps:

1) Problem representation in a format usable by GA
2) Computation of fitness value of the population
3) Cross-over between the selected population based on the fitness scores
4) Mutation of the generated population

7) (a) What is the major difference between Propositional Logic and First-Order Logic? (2 pts)

Whereas propositional logic is limited to facts (symbols), first order logic allows one to define relations and functions using these facts.

(b)Write the following sentences in First Order Logic: (4 pts)

Note: Any other valid variation is also acceptable

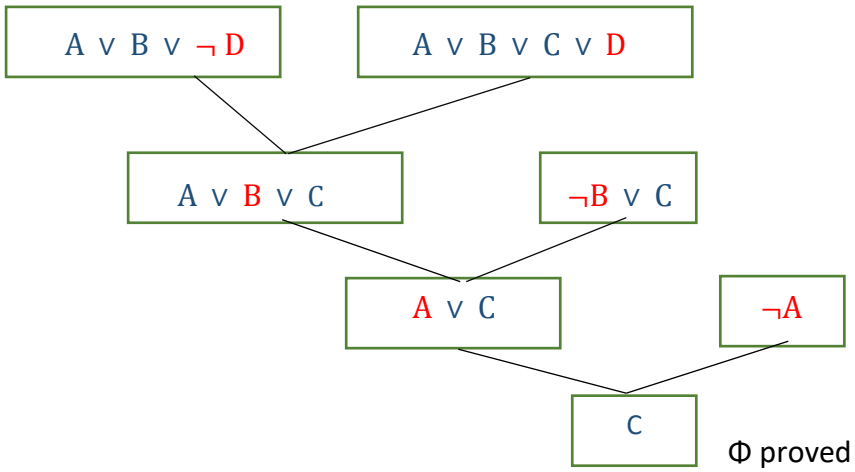
- Not every person is strong
  $\exists x\, \text{Person}(x) \wedge \neg \text{Strong}(x)$
- Every child has a mother
  $\forall x \exists y\, \text{Child}(x) \wedge \text{Lady(y)} \Rightarrow \text{Mother(y, x)}$

8) Give one disadvantage each of top-down and bottom-up parsing grammar in NLP? (2 pts)

Top-down parsing can get stuck in infinite loops while parsing due to grammar rules and also takes up a lot of time and space (listing any one shortcoming is enough). Bottom-up parsing finds it hard to backtrack to reparse once wrong parsing is done.

9) Prove Φ using Resolution, given the following knowledge base: (3 pts)

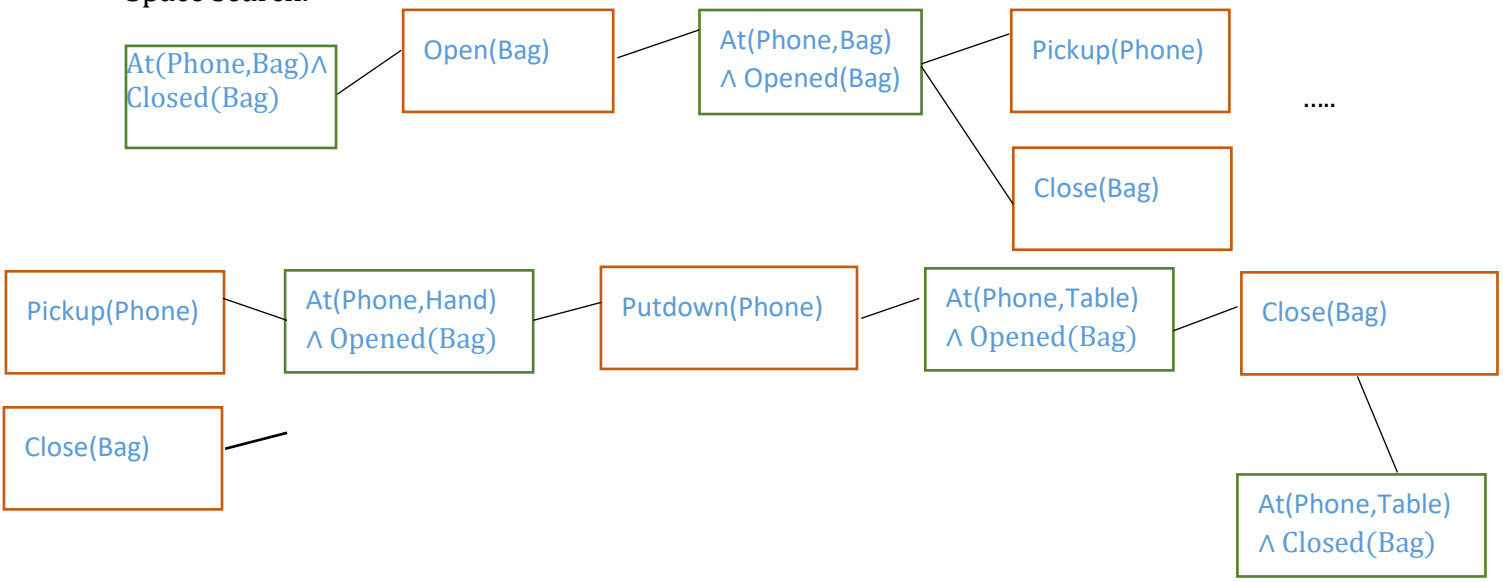
$\Phi_1 = A \vee B \vee \neg D$
$\Phi_2 = A \vee B \vee C \vee D$
$\Phi_3 = \neg B \vee C$
$\Phi_4 = \neg A$
$\Phi = C$



10) Given the following planning problem: (4 pts)

Init(At(Phone,Bag) ∧ Closed(Bag))
Goal(At(Phone,Table) ∧ Closed(Bag))
Action(Open(Bag)),
PRECOND: Closed(Bag),
EFFECT: Opened(Bag)
Action(Pickup(Phone)),
PRECOND: Opened(Bag),
EFFECT: At(Phone,Hand)
Action(Close(Bag)),
PRECOND: Opened(Bag),
EFFECT: Closed(Bag)
Action(Putdown(Phone)),
PRECOND: At(Phone,Hand),
EFFECT: At(Phone,Table)

Solve the planning problem using either the Forward State-Space or Backward-State-Space search.



Note: Here it was solved by Forward-State search method, backward-search method would be as acceptable. Further, you can actually first close the bag and then put the phone on table which is also a right solution. Any other valid variations would also be fine. Also note that actions and conditions are differentiated by boxes carrying different colors.